

Redefining Olympic Success

By: Ryan O'Loughlin and Christian Tuttle

The Context

Traditional Olympic medal counts heavily favor wealthy and highly populated nations. Our team noticed this inherent bias in global rankings.

The Environment

We set out to look beyond raw numbers and redefine athletic success through the lens of "**Olympic Efficiency**"—evaluating performance relative to economic and demographic constraints.

Data Challenges & Objectives

The Challenge: Transform an individual-level Olympic dataset into a structured country-level performance dataset.

Data Acquisition: Source data from website like Kaggle, united nations, world bank group, etc.

Aggregating Data/Feature Engineering: Combine Olympic dataset with datasets such as: GDP, Population, Human Development Index (Later On).

Missing Data: Experimented with different strategies such as using the mean for a feature. Dropping the data was the best outcome in most cases.

Normalization: Used MinMaxScaler to normalize features, ensuring consistent scale across variables.

The Engineering Pipeline

Our Model Creation Pipeline



1. Merging/Cleaning

Downloaded Individual Dataset,
transform it and merge it with
GDP/Population data.



2. Model Creation

Build multiple models, that use
different algorithms including
Regression, Random Forest,
XGBoost

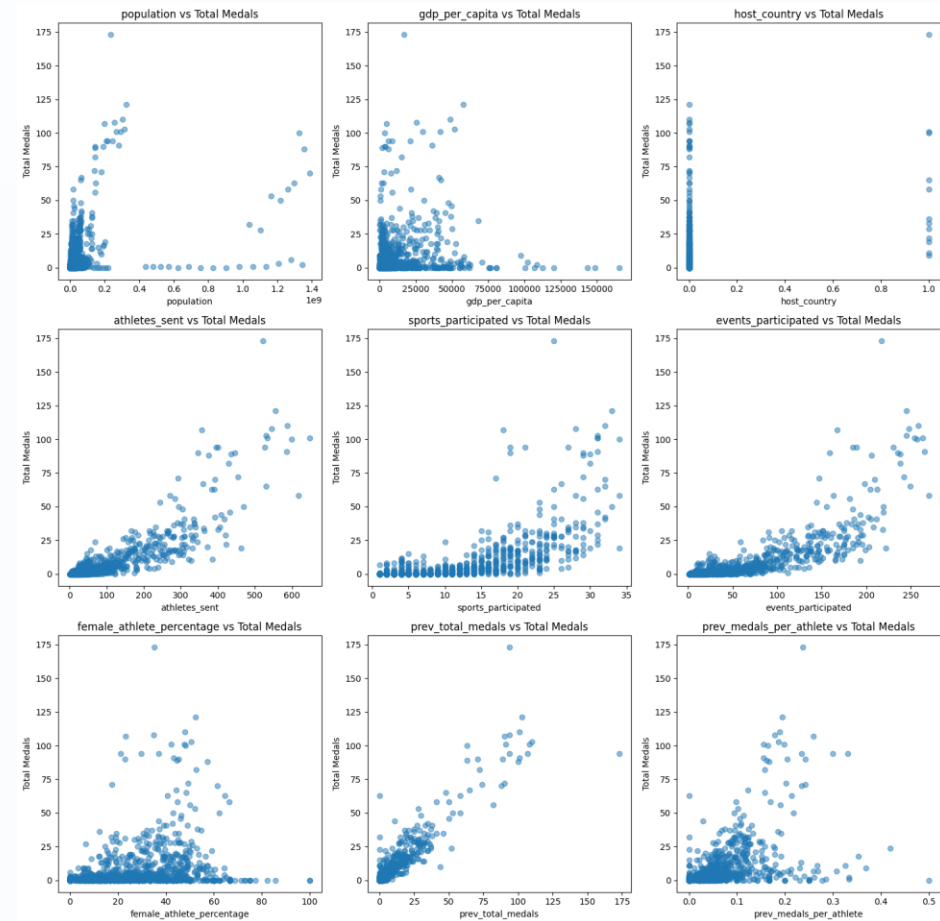
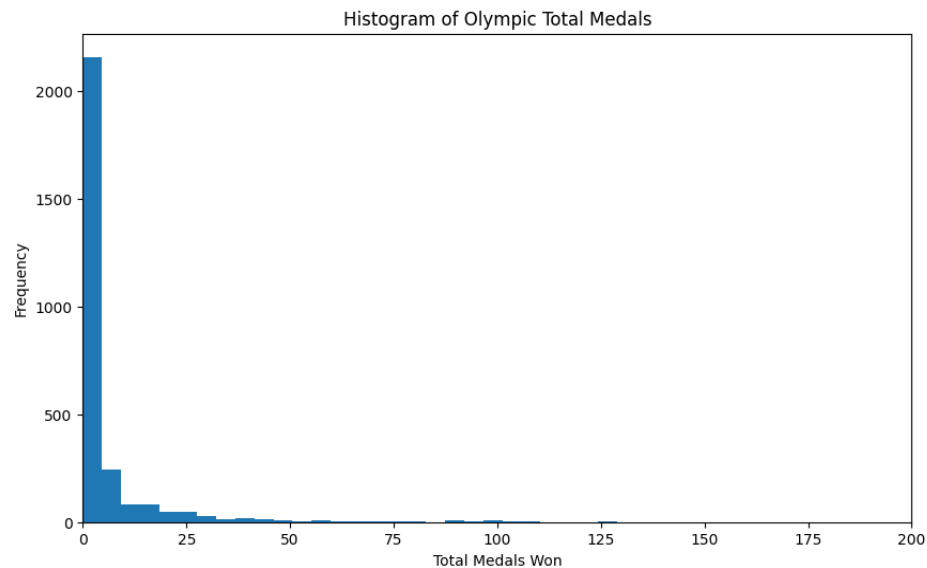


3. Model Evaluation

Test which models performed the
best and which features/variables
are the most important to the
model. Make adjustments from
there.

Visualization

- Unbalanced Dataset
- Most countries win very few medals
- Variables with high correlation:
 - Previous medals won
 - Athletes sent
 - Events participated



Our Prediction Model

Performance based model. We wanted to create a model that was the most accurate at predicting total medals for a country.

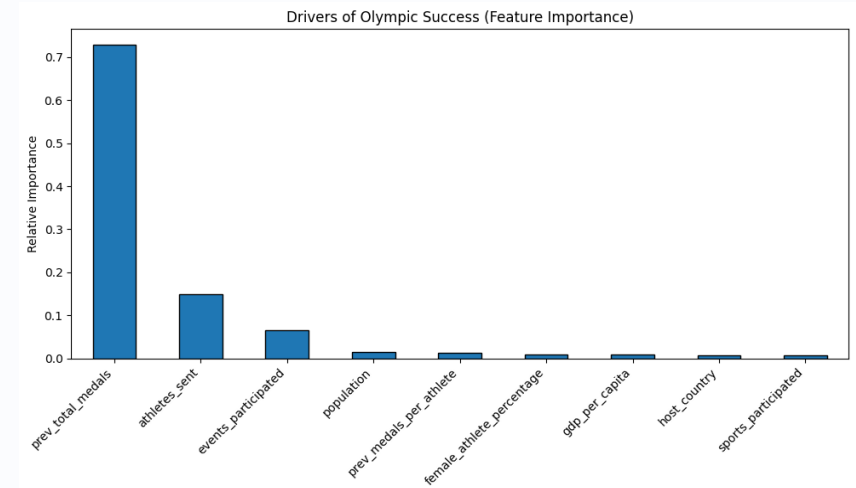
Most Important Features:

- Previous Medals Won
- Athletes Sent

Model Results (Using Cross Validation):

- **Linear Regression: 0.8933 R^2**
- **Random Forest: 0.9059 R^2 (Winner)**
- **XGBoost: 0.9013 R^2**

For the App, we use a model that only uses previous medals won and athletes sent for simplicity (Using Random Forest). This avoids overfitting.



Our Efficiency Model

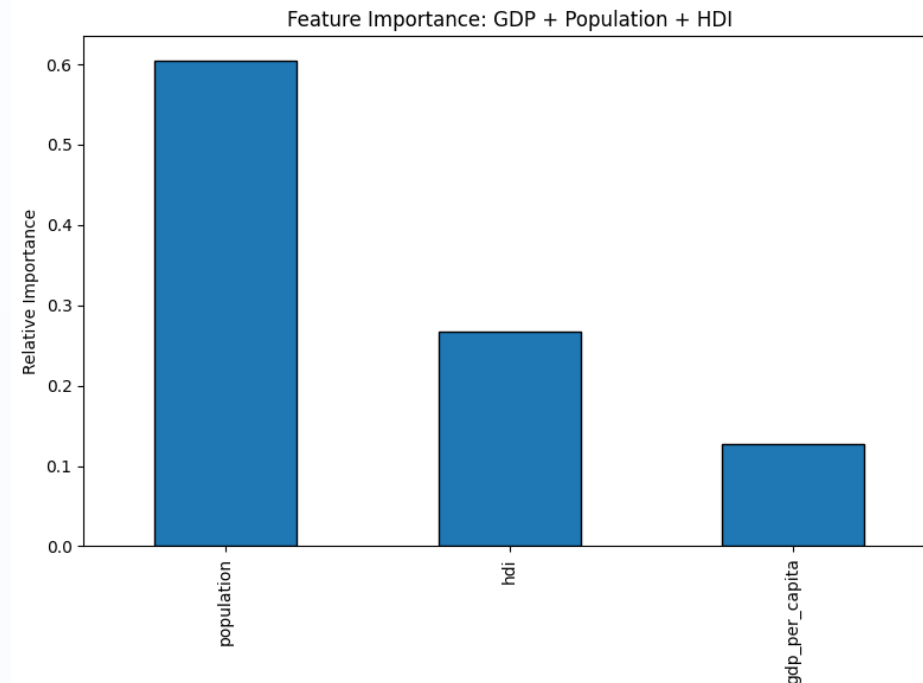
We wanted to understand how well a country's economic and development factors alone can explain Olympic success. And use that to decide how efficiently a country competes.

Features Used:

- Population
- GDP Per Capita
- HDI (Added to improve model)

Model Results (Using Cross Validation):

- **Linear Regression: 0.27 R^2**
- **Random Forest: 0.75 R^2 (Winner)**
- **XGBoost: 0.71 R^2**



Side Note:

A Neural Network was attempted...results were not nearly as accurate. Most likely due to small dataset size.

The New Olympic Leaderboard

The measurable outcome: Using our Random Forest predictions, we isolated countries that vastly over-perform relative to their resources.

Jamaica: One of the most efficient nations globally

Kenya: Massive outperformance in Long-Distance running.

USA/China: High absolute medals, but average efficiency when adjusted for GDP.



Our App

Created an app to host our model:

- Saved models to a file using pickle
- Created using Flask
- Deployed using Heroku (can visit the site from any device)
- [Link to our App!](#)